PREP XI RR

STUDY NOTES

ENT

19
VIDEOS

202
PAGES

PREMIUM STUDY MATERIAL

PREP XI RR

APRIL 2026

YOURWISH

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ENT

RAPID REVISION

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1. ANATOMY OF THE EAR

ANATOMY OF THE OUTER EAR

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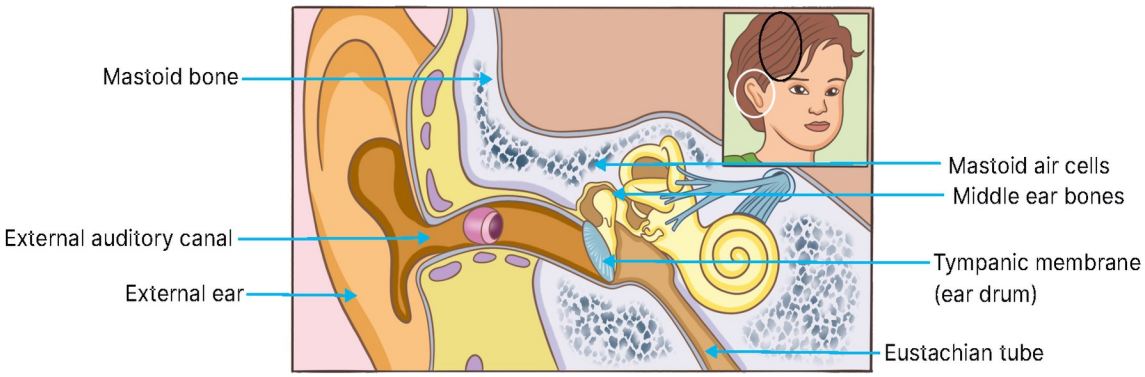
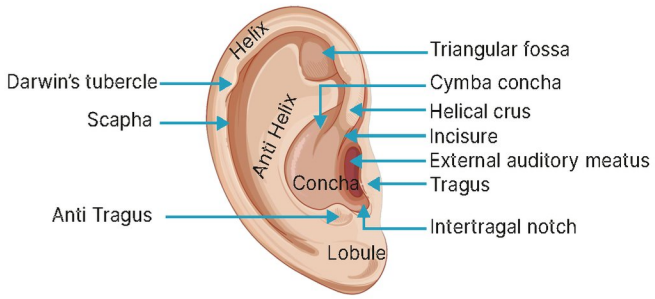


Ear anatomy and mastoid bone

- Components of the External Ear
  - Pinna.
  - External Auditory Canal (EAC).
  - Tympanic Membrane (TM)

Pinna

- Cymba conchae: cartilaginous landmark for mastoid antrum
- Sites devoid of cartilage:
  - Lobule: site to harvest fat graft for stapedectomy
  - Incisura terminalis: site of Lempert endaural incision
  - Darwin's tubercle: prominence at junction of upper ⅓ and lower ⅔ of helix



EAC

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CARTILAGENOUS EAC	BONY EAC
Outer 1/3rd (8mm)	Inner 2/3rd (16mm)
Completely developed at birth	Not developed at birth, continues to develop after birth
Directed upward, backward, and medially	Directed downward, forward, and medially
Contains thick skin (stratified squamous keratinized) with ceruminous (apocrine) glands and hair follicles	• Contains thin skin (stratified squamous keratinized) without skin appendages. • Has the narrowest part of EAC, called the isthmus, 5-6 mm lateral to TM
Has deficiency anteroinferiorly k/a fissures of santorini	Has a deficiency k/a foramen of Huschke

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- In neonates, since the bony part of the EAC is not developed, the pinna is drawn downward and backward to view the TM
- To straighten the EAC to visualize TM, the pinna should be pulled posterosuperiorly in adults
- The bony-cartilaginous junction, fissures of Santorini (deficiency in cartilaginous part), and foramen of Huschke (deficiency in bony part) are the potential paths for the spread of infections and tumors from EAC to the base of skull and parotid and vice versa

PARTS OF TYMPANIC MEMBRANE:

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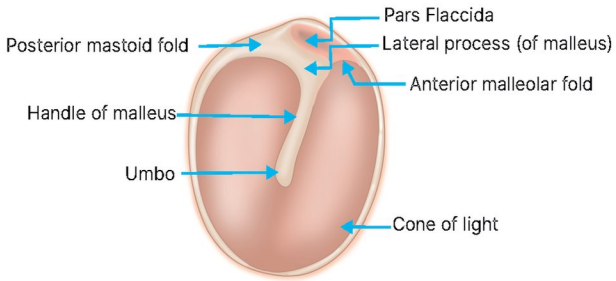
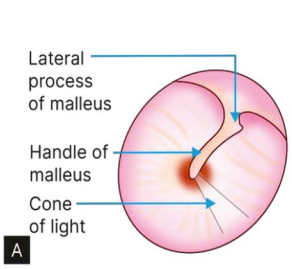
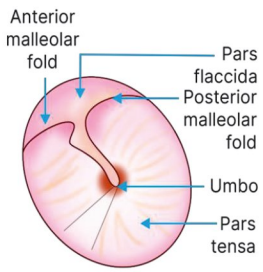


	Pars tensa	Pars flaccida
Annulus	Present	Absent
No. of layers	3	2
Umbo and cone of light	Umbo in the center and a cone of light in the antero-inferior part	

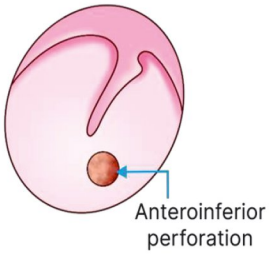
- Color: pearly gray and translucent
- Angle with EAC: 55 degrees
- Effective vibratory surface area is 55 mm^2.
- Total surface area: 90 mm^2
- Thickness: 0.1 mm
- Pars flaccida: also called Shrapnell's membrane
- Umbo: point of attachment of the tip of the handle of the malleus to the TM
- Side Identification (Right vs. Left)
  - Based on Cone of Light Position

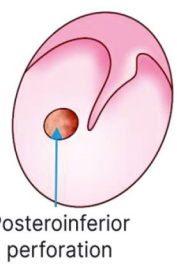




→ Right TM: Cone of light is between 3 to 6 o'clock.
→ Left TM: Cone of light is between 6 to 9 o'clock.

  - Based on Malleolar Folds





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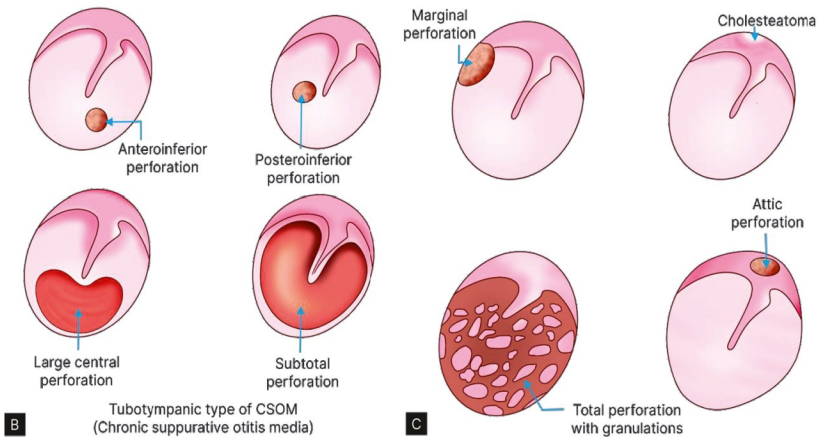
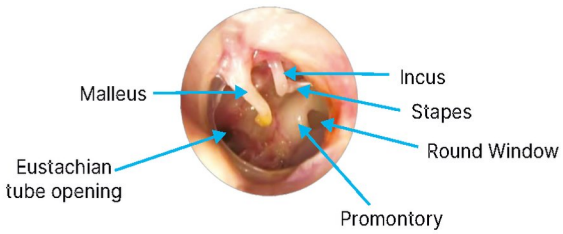
→ Short fold = Anterior malleolar fold.
→ Long fold = Posterior malleolar fold.

Perforations:

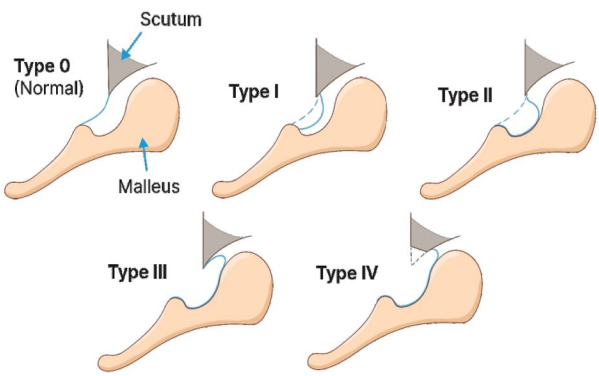


- Classification
  - Central Perforation: Located in the Pars Tensa.
  - Attic Perforation: Located in the Pars Flaccida.
  - Small Central: Occupies one quadrant of the Pars Tensa.
  - Large Central: Occupies more than one quadrant but not all; some Pars Tensa remains circumferentially.
  - Subtotal: Occupies all quadrants; Annulus remains intact.
  - Total: Occupies all quadrants; includes destruction of the Annulus.
  - Marginal: Occupies one quadrant and includes the Annulus.
- Structures Visible Through a Subtotal Perforation
  - Malleus: Specifically the handle.
  - Incus: Specifically the long process.
  - Stapes: Head, tendon, and crura.
  - Oval Window: Covered by the footplate of the stapes.
  - Round Window: Located inferior to the oval window.
  - Eustachian Tube: Opening seen antero-inferiorly.
  - Promontory: Projection on the medial wall formed by the basal turn of the cochlea.
  - Facial Nerve: Impression located above the oval window

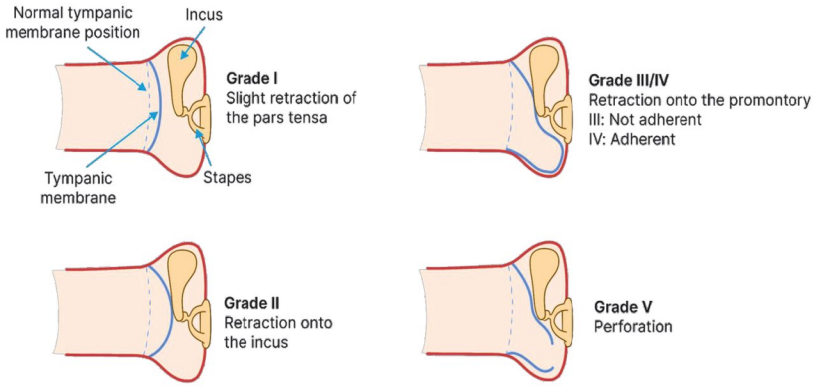


Classifications of retractions:

- Pars Flaccida: Toss Classification
  - Grade 0: Normal.
  - Grade 1: Small dimpling; does not touch the neck of the malleus.
  - Grade 2: Retracted to touch the neck of the malleus.
  - Grade 3: Adherent to the neck of the malleus and extending beyond the scutum (outer attic wall)
  - Grade 4: Retracted to the neck, extending beyond the scutum + Resorption of the outer attic wall



- Pars Tensa: Sade Classification
  - Grade 1: Membrane slightly retracted
  - Grade 2: Membrane touching the Incus
  - Grade 3: Membrane touching the Promontory but remains mobile
  - Grade 4: Membrane touching the Promontory, adherent and not mobile
  - Grade 5: Retraction with an atelectatic membrane + Perforation

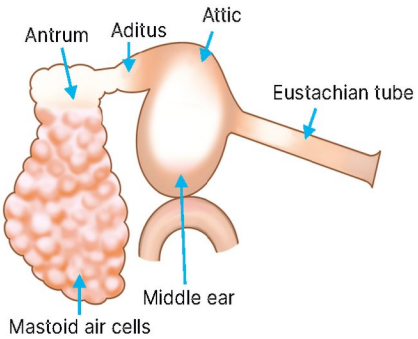


MIDDLE EAR

00:17:34

Middle Ear Cleft

- Air enters the middle ear via Eustachian tube → Middle ear proper → Attic → Aditus → Mastoid antrum



Middle Ear Anatomy:

00:18:04

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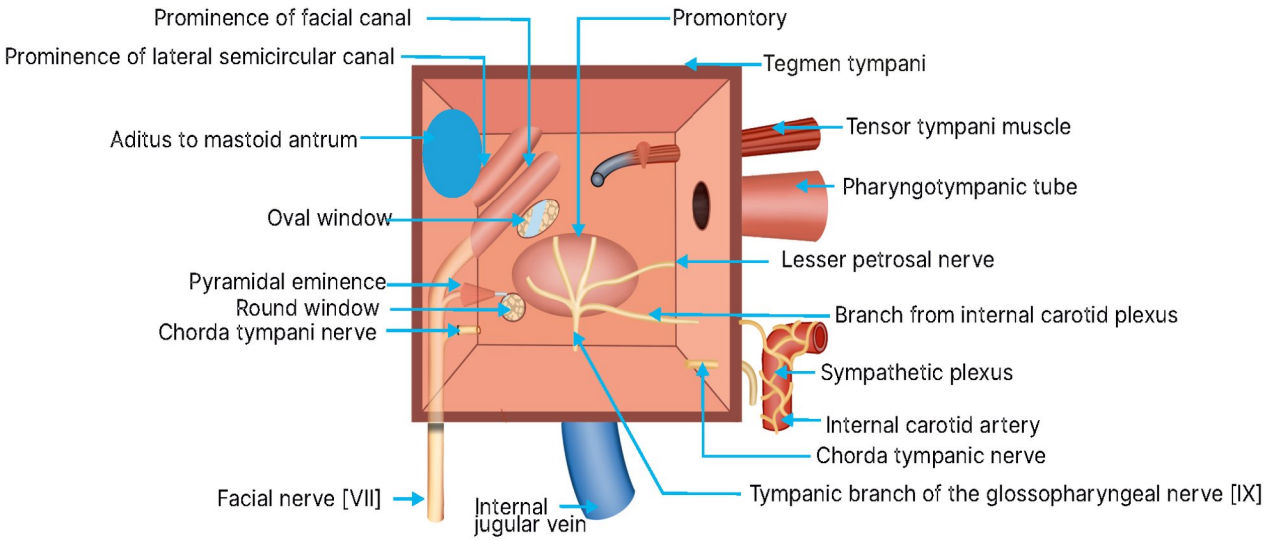

Roof	Tegmen tympani
Floor	Bone-separating jugular bulb
Anterior wall	• Communicates with ET • Bone-separating carotid • Communicates with canal for tensor tympani
Posterior wall	Pyramid and aditus
Lateral wall	TM and outer attic wall called scutum

Medial wall	• Impression of lateral semicircular canal • Promontory: impression of cochlea • Oval window • The facial canal passes between the semicircular canal and the promontory • Round window • Processus cochleariformis
Promontory	• Formed by the basal turn of the cochlea (senses high-frequency sounds)
Processus cochleariformis	• Landmark for the first genu of the facial and geniculate ganglion • The tensor tympani takes a lateral turn from here to attach to the upper end of the malleus handle
Facial nerve tympanic/horizontal/3rd segment	• MC dehiscent segment of the facial nerve
Lateral semicircular canal bulge	• MC SCC eroded/fistula formed by cholesteatoma. • Stimulated by the caloric test • Acts as a landmark for the tympanic segment as well as the second genu of the facial nerve.
Oval window	• The foot process of the stapes present here transmits sound vibrations to the scala vestibuli
Round window	• Covered by a secondary TM, the electrodes of a cochlear implant are inserted into the inner ear (S. tympani), and drugs are delivered to the inner ear through this window.
Tympanic plexus	• Formed by JACOBSEN'S NERVE and the sympathetic plexus around ICA (superior and inferior caroticotympanic nerves), it gives sensory supply to the middle ear, mastoid, TM, ET, and parasympathetic supply to the parotid through the lesser petrosal nerve (through the otic ganglion)

Recess	Boundaries	Clinical note
Facial recess	• Medial wall: vertical segment of the facial nerve • Lateral wall: chorda tympani nerve • Superior wall: fossa incudes	• Important surgical corridor (posterior tympanotomy) to access the middle ear without damaging ossicles • Posterior tympanotomy is used for Cochlear implant surgery
Sinus tympani	• Superior boundary: ponticulus (bony ridge between oval window and facial canal) • Inferior boundary: subiculum (bony ridge between the round window and the facial canal)	• Hidden recess behind the facial nerve → notorious site for residual cholesteatoma

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NERVE SUPPLY OF EAR

00:24:51

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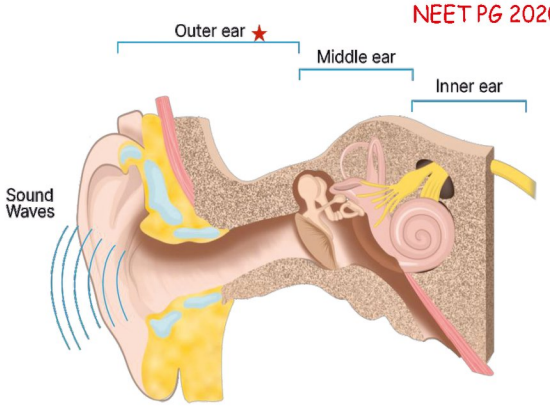
Outer Ear

- Lesser occipital nerve: a smaller portion of the pinna
- Auriculotemporal nerve: upper half anteriorly, roof, anterior TM
- Greater auricular nerve: lower half of the pinna
- Auricular branch of 10th nerve: Arnold's nerve, supplies the floor and posterior wall and the posterior TM



Middle ear:

- Supplied by Jacobson's nerve (Tympanic branch of Glossopharyngeal nerve)

Inner ear:

- Cochlea supplied by: cochlear nerve
- Vestibule supplied by: vestibular nerve

Intratympanic muscles and tympanic plexus:

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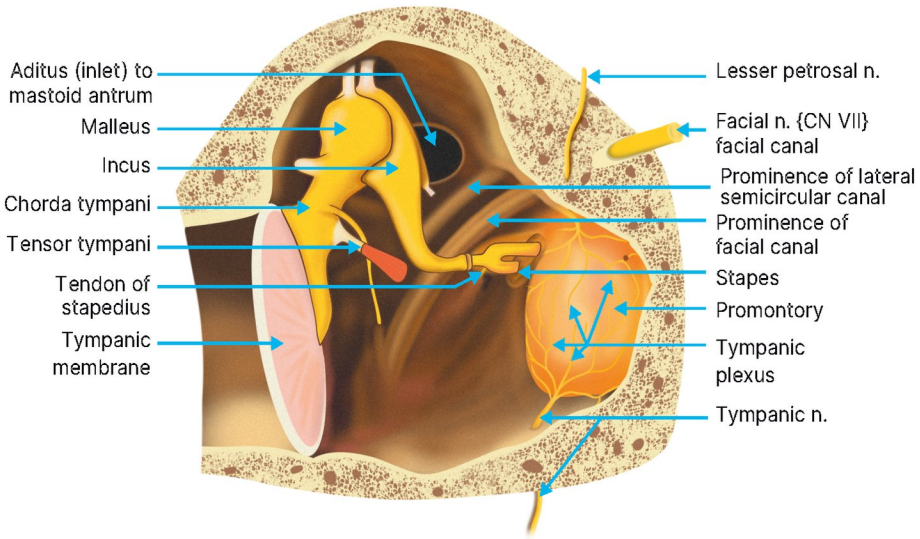

Tympanic Plexus

- Location: Situated on the promontory.
- Formed by the Glossopharyngeal nerve & Sympathetic plexus covering the carotid

PYQ:

Malleus:

- The largest ossicle of the middle ear
- The tensor tympani tendon inserts on the medial surface of the upper handle
- The chorda tympani passes just above the tendon insertion (posterior to the anterior wall)

Incus:

- The body articulates with malleus head → incudomalleolar joint (saddle-type synovial)
- Most common ossicle to undergo necrosis in otitis media → long process of incus
- The lenticular process is most vulnerable due to precarious blood supply

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Stapes:

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- The smallest ossicle in the entire body
- Articulates with the lenticular process of the incus → incudostapedial joint (ball-and-socket type)
- The stapedius tendon inserts on the posterior part of the neck
- The footplate lies on the oval window and transmits sound to the inner ear
- Footplate fixation seen in otosclerosis

ANATOMY OF INNER EAR

00:29:23

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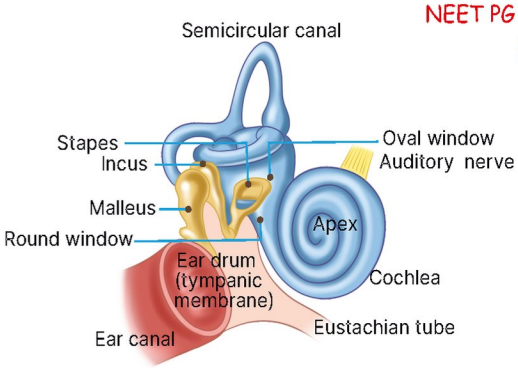
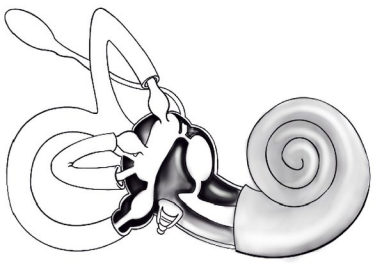
- Inner ear is aka labyrinth
- The central chamber is called the vestibule.
- Anteriorly → cochlea
- Posteriorly → semicircular canal



Vestibule:

- The spherical recess has a saccule.
- Elliptical recess has a utricle:
  - Receives the openings of ducts of the semi-circular canals
- Spherical and elliptical recess are parts of bony labyrinth, and saccule and utricle are parts of the membranous labyrinth
- Neurosensory epithelium within utricle and saccule is called maculae



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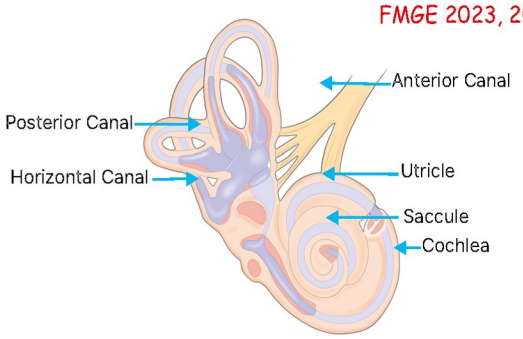
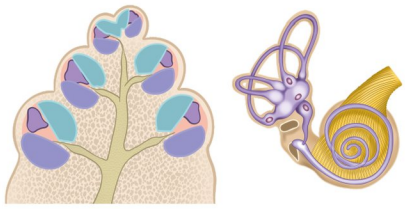
Semicircular canals:

- Responsible in maintaining body posture
- The canals have a dilated amputated end and a non-dilated end
- Crus communes: non-dilated end of superior and posterior canal joining together
- Within the amputated end, it has neurosensory epithelium and is called a "crista."



Cochlea:

- Spiral structure, has a central access called the modiolus, which has the cochlear artery and cochlear nerve
- The central chamber is called cochlear duct (scala media), between scala vestibuli and scala tympani (basilar membrane)
- Neurosensory epithelium above basilar membrane known as organ of Corti, which helps with hearing
- Neurosensory epithelium in scala media is striae vascularis which is responsible for production and absorption of endolymph



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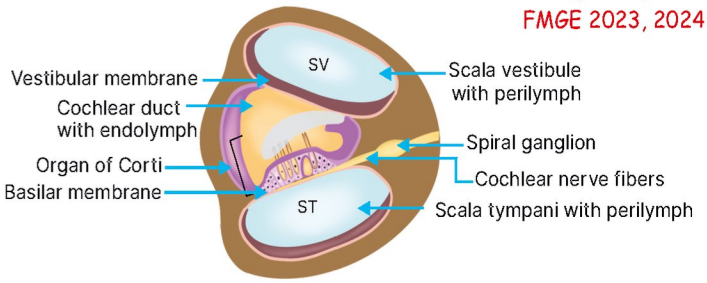
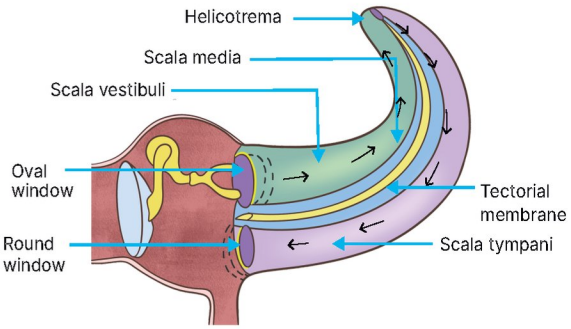
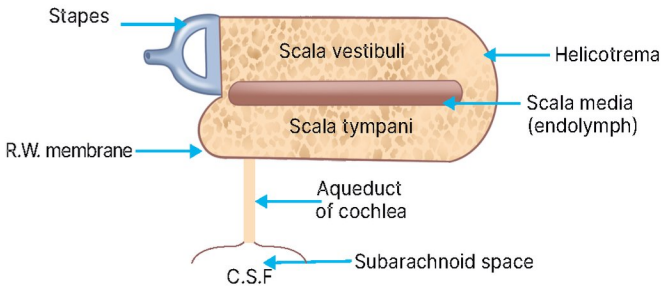


- The scala vestibuli and tympanic are bony labyrinths, and scala media is a membranous labyrinth
- In the base, scala vestibuli opens as oval window and scala tympani opens as round window
- At the apex, scala vestibuli and scala tympani communicate, known as the helicotrema
- Scala media: Situated in the middle, between the scala vestibuli and scala tympani.



Inner Ear Fluids: Production And Regulation

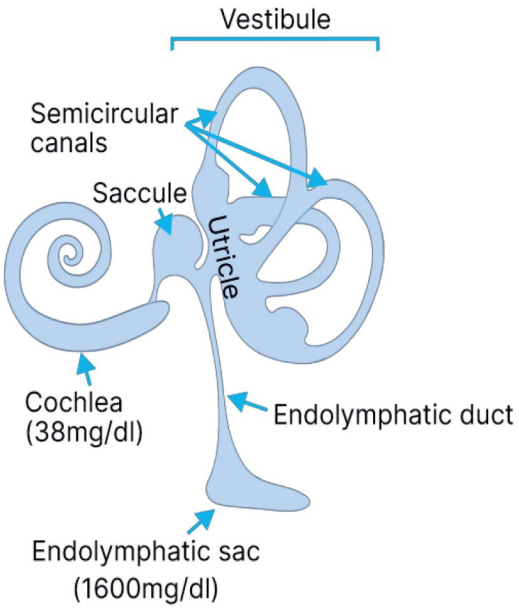
- Endolymph
  - Produced and reabsorbed by the stria vascularis.
  - Stria vascularis: Vascular epithelium located in the scala media.
  - Contains Sodium (Na+), Potassium (K+), and Chloride (2Cl-) channels for regulation.
  - Diuretic usage → affects Na+/K+/2Cl- channels → hearing loss and vertigo



- Perilymph
  - Composition: An ultra-filtrate of Cerebrospinal Fluid (CSF).
  - Flow: Originates from CSF → enters scala tympani → fills space between membranous and bony labyrinths.
  - Cochlear aqueduct: The duct facilitating communication between the scala tympani and the subarachnoid space.
  - Distribution: Endolymph is inside the membranous labyrinth; perilymph is between the membranous and bony coverings.

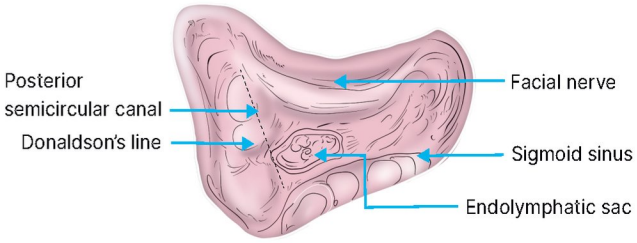
Endolymphatic Duct

- Vestibule (Central Chamber)
  - Bony part: Contains spherical and elliptical recesses.
  - Membranous part: Contains the saccule and utricle.
- Saccular duct + Utricular duct → Endolymphatic duct.
- Endolymphatic duct → Opens into the endolymphatic sac.
- Utricle: Receives the openings of the semicircular canal ducts via five openings.
- Saccule: Communicates anteriorly with the cochlea via the cochlear duct (scala media).
- Bony Coverings of Membranous Structures
  - Cochlear duct: Covered by scala vestibuli (above) and scala tympani (below).
  - Saccule: Covered by the spherical recess.
  - Utricle: Covered by the elliptical recess.
  - Semicircular canal ducts: Covered by the semicircular canals.
  - Endolymphatic duct: Covered by the vestibular aqueduct



Surface landmarkings:

- Donaldson's line: an imaginary line drawn from the posterior semicircular canal that bisects the lateral semicircular canal into 2 halves.



EMBRYOLOGY

00:37:38

Ear part	Embryonic layer	Arch
Pinna	Mesoderm	1 and 2 arch
EAC	Ectoderm	1st cleft
Tympanic membrane	All three layers	1st cleft and 1st pouch along with mesoderm
Middle ear cleft	Endoderm	1st and 2nd forms tubotympanic recess
Malleus and incus	Mesoderm	1st arch
Suprastructure of stapes	Mesoderm	2nd arch
Footplate of stapes	Mesoderm	Otic capsule
Mastoid	Mesoderm	Squamous and petrous parts of temporal bone

ONE-LINERS

00:38:43

- The part of the pinna lying between the ascending crest of the helix and tragus is called incisura terminalis
- The major part of the skin of the pinna is supplied by the greater auricular nerve
- Arnold's nerve is a branch of the vagus nerve.
- In newborns, the bony part of the EAC is absent
- Fissure of santorini are seen in the cartilaginous part of the EAC
- The cough response caused while cleaning the ear canal is mediated by stimulation of the vagus nerve
- The tympanic membrane forms an angle of 55 degrees with the EAC
- Körner's septum is a persistence of the petrosquamous suture
- The space between the pars flaccida and neck of the malleus is called Prussak's space.
- The part of the middle ear near the Eustachian tube is called the protympanum
- The site of exit of the chorda tympani from the middle ear is called the canal of Hugier.
- The scutum is the bony part of the outer attic wall
- A patulous eustachian tube is commonly seen in obese patients
- The electrodes of a cochlear implant are placed into the scala tympani via round window
- Endolymph is present in the membranous labyrinth
- The eustachian tube is typically 36 mm long
- The structure responsible for endolymph is the striae vascularis
- Duct of cochlea connecting with subarachnoid space → Aqueduct of cochlea
- The endolymphatic duct is formed by Saccular duct + Utricular duct
- The endolymphatic duct is covered by vestibular aqueduct
- Cochlear duct is → Scala media
- The cochlear duct is connected to the saccule by the ductus reunions

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- The scala media is separated from scala tympani by basilar membrane
- The scala media is separated from the scala vestibuli by Reissner's membrane.

PYQs

00:42:58

Q. What is the function of the utricle and saccule?

A. Hearing high-frequency sounds

B. Linear acceleration and deceleration

C. Angular acceleration

D. Hearing low-frequency sound

Answer: B

Q. Which of the following structures in the inner ear is responsible for balance? FMGE Jan 2024

A. Semicircular canal

B. Cochlea

C. Ampulla

D. Organ of corti

Answer: A

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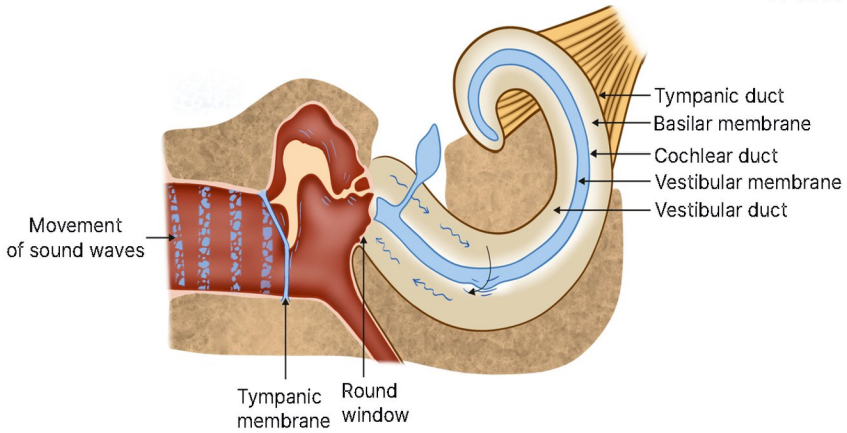
2. PHYSIOLOGY OF HEARING

CONDUCTIVE PATHWAY

00:01:00

Path Of Sound (air Conduction)

1. External auditory canal
2. Tympanic membrane
3. Malleus
4. Incus
5. Stapes (footplate)
6. Oval window



Sound Transmission In The Inner Ear

- Oval window → Scala vestibuli
- Scala vestibuli → Scala tympani (through inner ear fluid)
- Ends at the Round window

Phase Difference (space Difference)

- Sound hits the Oval window first, the round window later
- This phase difference is necessary (TIME DIFFERENCE)
- If both windows vibrate simultaneously
  - No phase difference
  - No proper mechanical vibration
  - No stimulation of the Organ of Corti
  - No conversion into electrical energy

SENSORY NEURAL PATHWAY

- Begins from the cochlea onward
- Beyond the middle ear = SENSORY NEURAL PATHWAY

Impedance Matching

00:03:00

- Middle ear = Air
- Inner ear = Fluid (Perilymph + Endolymph)
- Sound moving from air → fluid faces resistance
- The impedance matching mechanism helps overcome this resistance

Impedance Matching Mechanisms

Area Ratio (areal Ratio)

- Condensation of sound travelling from a larger tympanic membrane to a smaller foot plate of the stapes

Parameter	Value
Effective vibratory area of the tympanic membrane	45 mm^2
Stapes footplate area	3.2 mm^2
Calculation	45 ÷ 3.2 = 14:1
If vibratory area = 55 mm^2	55 ÷ 3.2 = 17:1
Area Ratio Range	14-17 : 1

Lever Ratio

- Condensation of sound travelling from a larger malleus to a smaller incus

Parameter	Value
Mechanism	Malleus (larger) → Incus (smaller)
Lever Ratio	1.3 : 1

Total Transformer Ratio

- Total transformer ratio= Areal ratio x Lever ratio

Parameter	Value
Effective vibratory area	45 mm^2 /55 mm^2
Foot plate area	3.2 mm^2
Areal ratio	14:1/17:1
Lever ratio	1.3:1
Total transformer ratio	14 x 1.3= 18:1 17 x 1.3= 22:1

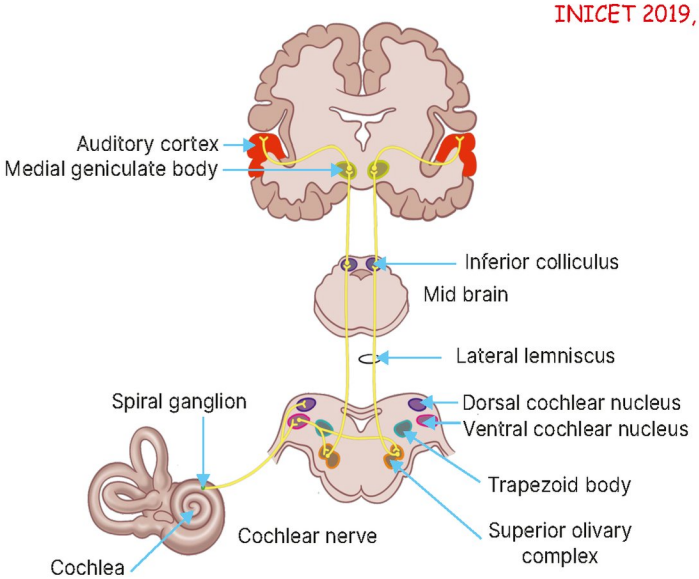
AUDITORY PATHWAY

00:06:00

INICET 2019, 2020, 2022, 2023

(Mnemonic: ECO LIMA)

- E- Eighth nerve
- C- Cochlear nucleus
- O- Superior olivary complex
- L- Lateral lemniscus
- I- Inferior colliculus
- M- Medial geniculate body
- A - Auditory cortex



Functions Of Different Levels

INICET 2019, 2020, 2022, 2023 00:08:00

Function	Site	Key Details
Frequency (Pitch) Determination	Cochlea	Tonotopic organization present
Amplitude (Intensity) Detection	Cochlear nerve	Detects the loudness of sound
Feature Detection	Higher auditory centers	Recognizing the type of sound (human, animal), human voice discrimination
Localization of Sound	Higher auditory centers	Detecting direction (right/left/front/back)

- **Tonotopic organization-** The base of the Cochlea is responsible for recognizing high-frequency sounds, but the apex of the cochlea is responsible for recognizing low-frequency sounds
- **Retrochochlear pathway defect-** Speech discrimination is affected; unable to identify phonetically similar (E.g. Fat, Bat, Rat and Fa, Ba, Ma sound similar)

TRANSDUCTION & HAIR CELLS

00:10:00

Transduction

- Conversion of Mechanical energy → Electrical energy
- Location: Organ of Corti

Hair Cells

Type 1 Cells (Inner Hair Cells)	Type 2 Cells (Outer Hair Cells)
Afferent function	Efferent function
True sensory receptors	Control Type 1 activity
Secrete Glutamate (Excitatory neurotransmitter)	Secrete GABA (Inhibitory neurotransmitter)

Supporting Cells

- Cells of Deiters
- Cells of Hensen
- Function: Support cells of the Organ of Corti

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3. ASSESSMENT OF HEARING

TUNING FORK TESTS

00:01:40

Air Conduction And Bone Conduction

- Air Conduction (AC)
  - Tuning fork placed in front of pinna.
  - Pathway: External Auditory Canal → Tympanic Membrane → Malleus → Incus → Stapes → Inner Ear (Organ of Corti)
  - Organ of Corti converts mechanical energy to electrical energy → 8th Nerve → Brain
  - AC = Conductive Pathway + Sensory Neural Pathway.
- Bone Conduction (BC)
  - Tuning fork placed on mastoid bone.
  - Pathway: Vibrations go directly to cochlea → Brain
  - Bypasses external auditory canal and middle ear.
  - BC = Sensory Neural Pathway alone

Rinne Test

AIIMS 2020, FMGE 2020, 2021, INICET 2022, 2023
NEET PG 2023

Rinne Positive	• AC > BC ○ Sound heard longer and louder via air conduction • Inference: Seen in Normal hearing or SNHL (Sensorineural Hearing Loss)
Rinne Negative	• BC > AC ○ Bone conduction is heard longer and louder than air conduction • Inference: Seen in Conductive Hearing Loss
False Negative Rinne	• Mechanism: Trans-cranial transmission of sound • Inference: Seen in Unilateral Severe to Profound SNHL (Dead ear)

Rinne Test And Degree of Hearing Loss

256 Hz	512 Hz	1024 Hz	Degree of Hearing Loss
-	+	+	Mild (15-30 dB)
-	-	+	Moderate (30-45 dB)
-	-	-	Severe (>45 dB)

Weber Test

- Method: Tuning fork placed in midline.
- Normal: Sound distributed equally to both ears.
- Lateralization:
  - Sound goes towards Diseased Ear → Conductive Hearing Loss.
  - Sound goes towards Normal/Opposite Ear → SNHL.

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- Example: Lateralization to Right ear implies:
  - Right Conductive Hearing Loss OR
  - Left SNHL.
- **Note:** Must combine with Rinne and Absolute Bone Conduction tests for diagnosis.

Absolute Bone Conduction (abc) Test

- Tests Sensory Neural Pathway (identifies SNHL)
- **Method:**
  - Patient and Examiner occlude tragus (Test assumes examiner has normal hearing sensitivity)
  - Compare BC duration of patient vs. examiner
- **Inference:**
  - **Normal:** Patient and Examiner hear for same duration
  - **SNHL:** Patient hears for shorter duration than examiner (ABC shortened)

PURE TONE AUDIOMETRY

00:11:43

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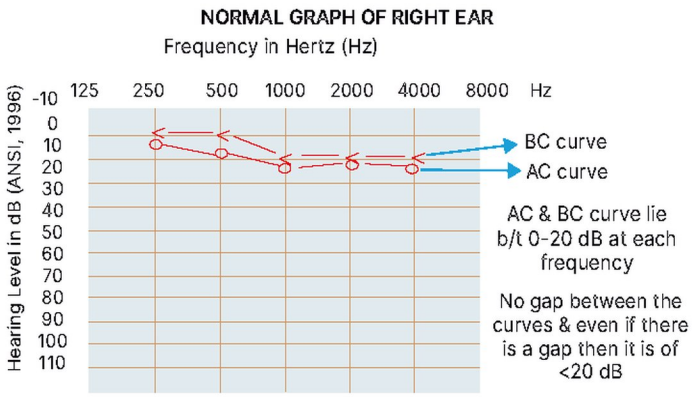
- Type of Test: Subjective test (Patient gives response)
  - Note: Not an objective test despite using a machine
- Test is Conducted in a soundproof room
- AC and BC are tested separately from higher Db to a lower Db
- Lowest threshold is marked on the Audiogram

Symbols Used In Audiogram

	Right Ear	Left Ear
AC unmasked	O	X
AC masked	Δ	[]
BC unmasked	<	>
BC masked	[	]
No responses		

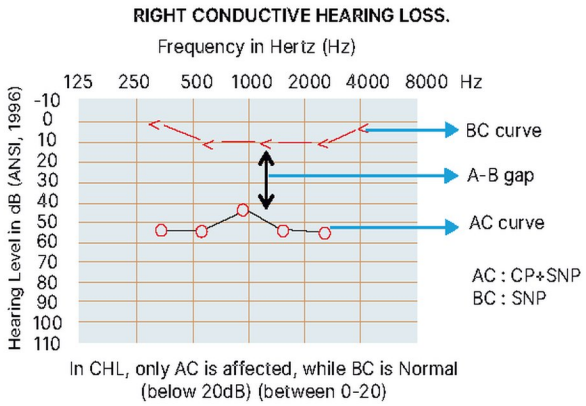
- R-ear depicted in red colour and Left-ear in blue color

Interpreting The Audiogram



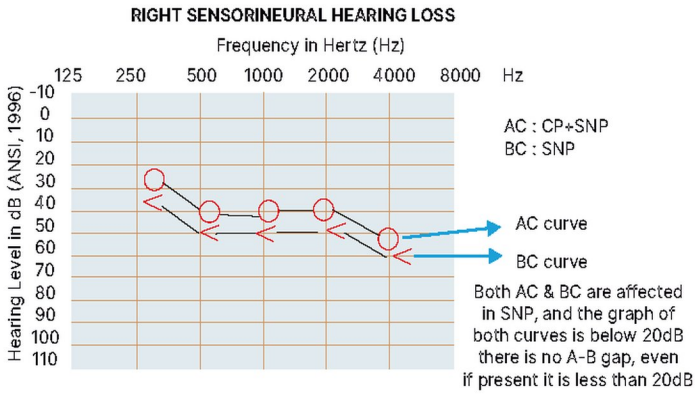
- Normal Hearing:
  - AC and BC curves lie between 0 to 20 dB.
  - No Air-Bone (AB) gap (or gap < 20 dB)

Conductive Hearing Loss



- BC Curve: Normal range (0-20 dB)
- AC Curve: Abnormal (Below normal range)
- AB Gap: Present (> 20 dB)
- Reason: Conductive pathway problem affects AC but spares BC (direct cochlear stimulation)

Sensorineural Hearing Loss (snhl)

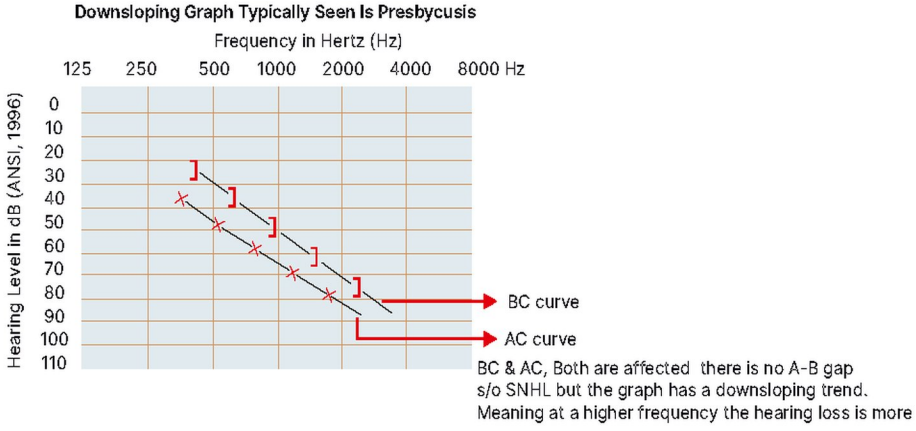


- AC Curve: Abnormal
- BC Curve: Abnormal
- AB Gap: Absent (Gap < 20 dB)
- Reason: Both AC and BC involve the sensory neural pathway, which is damaged.

Mixed Hearing Loss

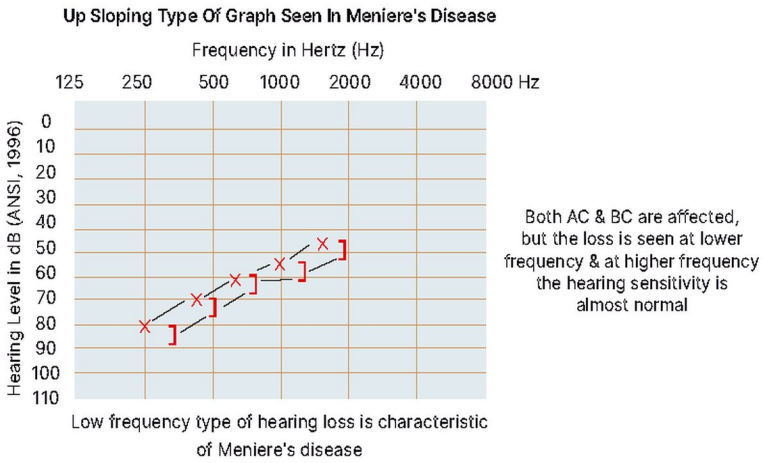
- AC Curve: Abnormal
- BC Curve: Abnormal
- AB Gap: Present (> 20 dB)
- Reason: Combination of conductive and sensory neural components.

Presbycusis



- **Definition:** Age-related Sensorineural Hearing Loss (SNHL).
- **Audiogram Findings:**
  - AC (Air Conduction) is abnormal
  - BC (Bone Conduction) is abnormal.
  - Neither curve is in the 0-20 dB range
  - Down-sloping Graph Pattern → Typical of presbycusis
  - Hearing loss worsens as frequencies increase.

Meniere's Disease



- **Pathology:** Inner ear condition causing SNHL.
- **Audiogram Findings:**
  - AC abnormal; BC abnormal
  - Up-sloping graph Pattern
  - More hearing loss at lower frequencies

Otosclerosis

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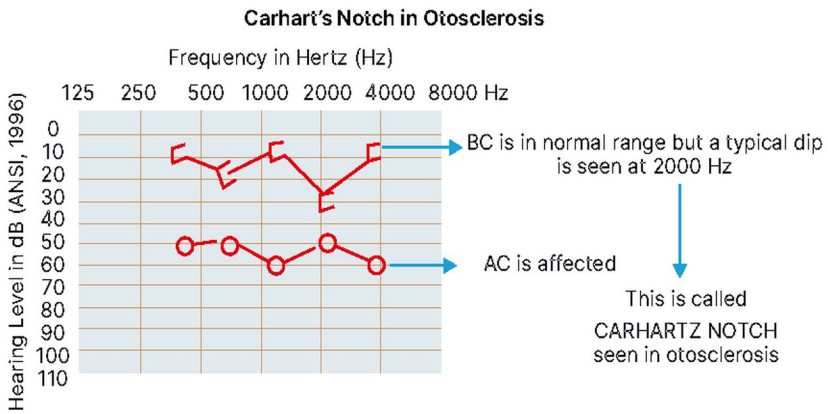

- **Pathology:** Conductive pathway problem → Conductive Hearing Loss (CHL).

Audiogram Findings

- BC is normal (0-20 dB range)
- AC is abnormal
- Air-Bone (AB) gap is present
- Specific Feature: Carhart's Notch.
  - A specific dip in the BC curve at 2000 Hz

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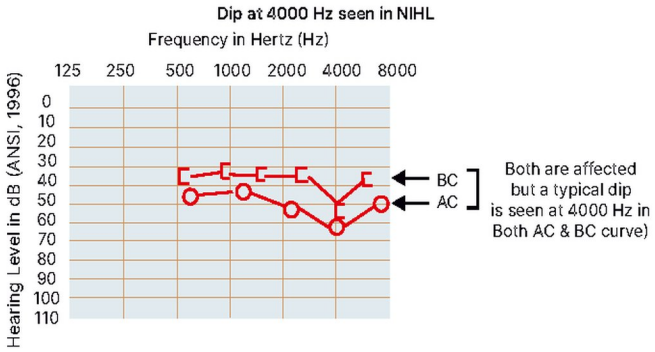
Noise-induced Hearing Loss (nihl)



- **Pathology:** Noise causes damage → SNHL.

Audiogram Findings

- AC and BC are both abnormal
- Specific Feature: Boiler's Notch / Acoustic Dip.
  - A dip seen in both AC and BC curves at 4000 Hz.

Summary Table

Types of hearing loss / disease	PTA findings
Conductive hearing loss	BC normal; AC abnormal; AB gap present (>15 dB)
Otosclerosis	AB gap + dip at 2k in BC (Carhart's notch)
Sensorineural hearing loss	Both BC and AC abnormal and AB gap <15 dB
Presbycusis	No AB gap / <15 dB; Down sloping audiogram (high frequency hearing loss)
Ototoxicity	Down sloping audiogram (high frequency hearing loss)
Noise-induced hearing loss	Dip at 4k (acoustic dip) in both AC and BC
Early Meniere's	No AB gap + Upsloping audiogram (low frequency HL)
Cochlear otosclerosis	No AB gap + U shaped / trough / cookie bite audiogram
Mixed hearing loss	Both BC and AC abnormal; AB gap present (>15 dB)

IMPEDANCE AUDIOMETRY

00:23:04

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- Components: Includes two tests:
  1. Tympanometry.
  2. Stapedial Reflex

Tympanometry

- Test done to assess middle ear pathologies
- Test Type: Objective test (Machine gives response, not patient)
- Parameters Measured:
  - Middle ear pressure
  - Mobility of the Tympanic Membrane (TM).

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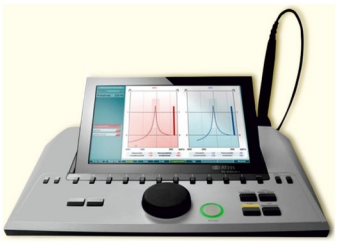
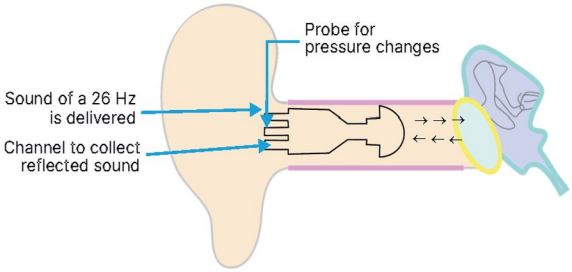
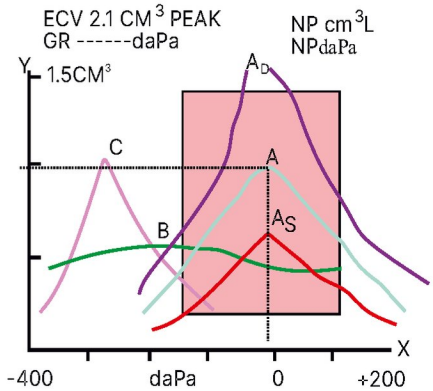
- Mechanism:
  - Probe placed in canal
  - Pressure in canal is ↑ /↓ to match middle ear pressure
  - TM vibrates at maximum when Canal Pressure = Middle Ear Pressure





Tympanogram Interpretation (graphs)



- Type A Graph:
  - Ascending and descending curve with peak at 0
  - Height: Peak occupies approx. half the length of Y-axis
  - Seen in the Normal middle ear.
- Type As Graph:
  - Peak at 0, but height is less than half of Y-axis
  - Seen in Otosclerosis
- Type Ad Graph:
  - Peak at 0 (or near it) but very high/open (off the chart)
  - Seen in Ossicular Discontinuity (e.g., following head injury/trauma).
- Type B Graph:
  - Single flat line (no peak)
  - Seen when Fluid or Effusion in middle ear (OME).
- Type C Graph:
  - Curve with peak shifted to negative pressure (-200, -300)
  - Seen in Eustachian Tube Dysfunction (Negative middle ear pressure).

Stapedial Reflex

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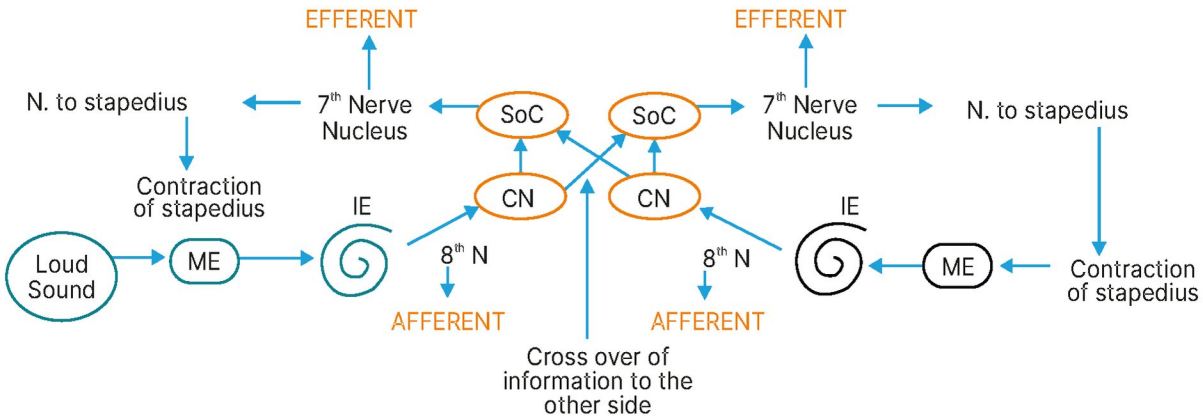
- Definition: Loud sound to one ear causes bilateral contraction of stapedius muscles
- Reflex Pathway
  - Loud sound → External/Middle/Inner Ear → Via 8th Nerve goes to Cochlear Nucleus → Ipsilateral and Contralateral Superior Olivary Complex (Note: Lateral decussation occurs here) → Stimulates B/l Facial Nerve Nucleus → Nerve to Stapedius (Branch of 7th nerve) → Contraction of Stapedius muscle



Important Points To Remember

Stapedius Reflex	Important Stapedius Reflex Results
• Afferent nerve – 8th CN • Center for stapedial reflex – Superior Olivary complex • Efferent nerve – 7th CN • Test which nerves – 8th and 7th • Giving a loud sound in the right ear → Rt. Ear 8th and 7th and left ear 7th • Bilateral reflex • 7th nerve palsy of right side → Reflex absent on Rt Side → Hyperacusis (normal sounds appear abnormally loud)	• CHL (otosclerosis, SOM) – absent • Sensory/Cochlear HL – threshold reduced (will be present at lower loudness due to recruitment) • Neural/retrocochlear HL – stapedial reflex decay present (the reflex will die out quickly inspite of continuous loud sound). • Malingers – Normal • Evaluating hearing in infants/young children unable to undergo tuning fork/audiometry

SPEECH AUDIOMETRY

00:32:30

- Two tests → Speech Reception Threshold (SRT) and Speech Discrimination Score (SDS)

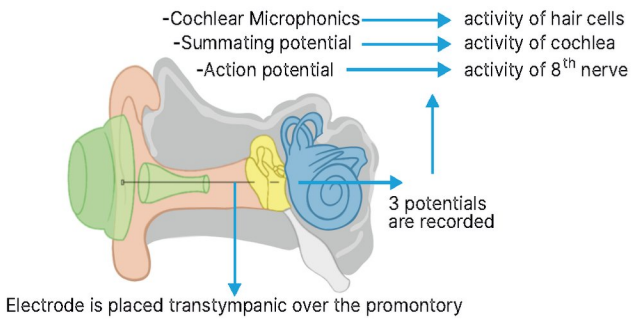
Speech Reception Threshold (SRT)	Speech Discrimination Score (SDS)
• It is a threshold at which 50% of spoken words are repeated correctly	• It is the ability of a patient to correctly identify phonetically balanced words like fox, box. • SDS < 20% strongly suggests retrocochlear pathology

- Both SRT and SDS are affected in Retrocochlear pathology

ELECTROCOCHLEOGRAPHY (ECOG)

00:35:20

- Type: Objective test for inner ear assessment
- Parameters: Measures inner ear potentials:
  - Cochlear microphonics.
  - Summating potential (SP).
  - Action potential (AP).
- Method: Electrode placed via tympanic membrane over the promontory (basal turn of cochlea).
- Interpretation:
  - Normal SP/AP ratio: < 30%.
  - Abnormal ratio (> 30%) → Indicates Meniere's disease.
- IOC for identifying Meniere's disease.



OTOACOUSTIC EMISSIONS (OAE)

00:36:12

- Emissions coming from Outer Hair Cells of the cochlea
- Significance:
  - Best audiometric screening test for neonates
  - Features: Non-invasive, repeatable, easily applicable, cost-effective
  - Screening investigation of choice in Neonatal Hearing Screening Program.
- Uses:

- Differentiate Cochlear vs. Retrocochlear pathology
- Early detection of Noise-Induced Hearing Loss (NIHL) in high-risk professions (factory, airline)

- Testing done in awake babies using a probe in the external auditory canal

Oae Result Interpretation

- OAE is Present ("Pass")
  - Outer hair cells/Cochlea are normal.
  - If SNHL persists → Likely Retrocochlear pathology
- OAE is Absent ("Refer")
  - Indicates abnormality (Cochlear pathology OR Middle ear pathology blocking emissions)
  - Does not confirm hearing loss; requires further testing

Management Protocol For "refer" (absent Oae)

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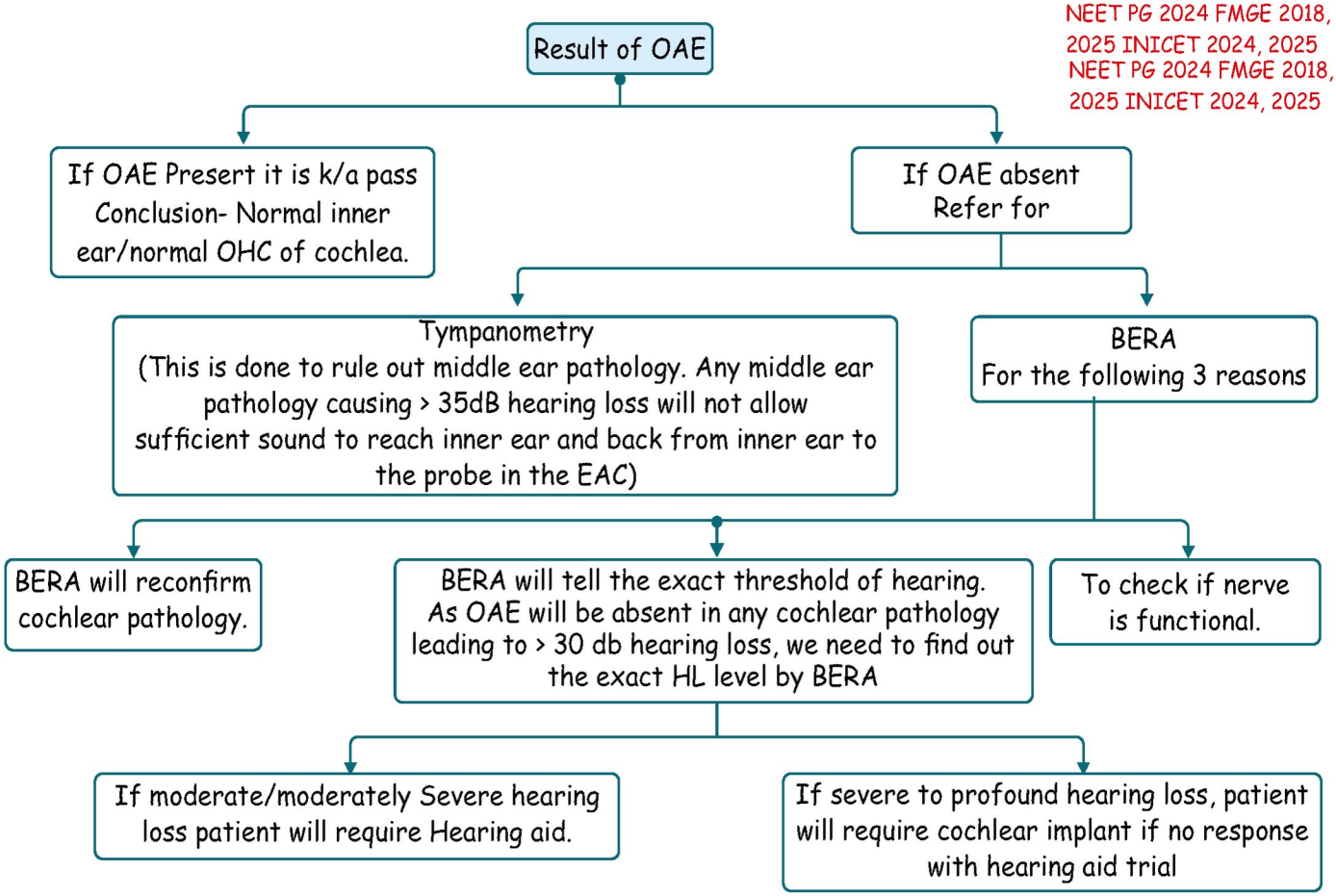

BERA (BRAINSTEM EVOKED RESPONSE AUDIOMETRY)

00:40:46

- Typically done in a sleeping child
- Done to Assesses auditory pathway from 8th nerve up to brain stem
- Type: Non-invasive objective test
- Wave Generation: 5 to 7 waves generated; each represents a part of the pathway.

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BERA Waves

Mnemonic: E-E-C-O-L-A-I

Wave	Structure Represented	Mnemonic Letter
Wave I	Distal part of CN VIII	E
Wave II	Proximal part of CN VIII near the brainstem	F
Wave III	Cochlear nucleus	C
Wave IV	Superior olivary complex	O
Wave V	Lateral lemniscus	L
Wave VI	Inferior colliculus	I
Wave VII	Inferior colliculus	I

Clinical Uses of BERA

Uses of BERA	• It is the best audiometric test • To screen hearing loss in neonates in ICU • To confirm hearing loss in neonates and infants To find hearing threshold • Differentiate between cochlear and retrocochlear HL • For retrocochlear pathology - acoustic neuroma • For nonorganic hearing loss (malingering)
BERA	Measures electrical activity in the auditory pathway from sound stimulus to 10 m/sec
MLR (Middle latency response)	Measures electrical activity in the auditory pathway 10-50 m/sec after sound stimulus
LLR (Late latency response / Cortical responses)	Measures electrical activity in the auditory pathway 50-500 m/sec after sound stimulus - tested by CERA (Cortical Evoked Response Audiometry)

- Best test for
  - Confirmatory test for hearing loss
  - Non organic hearing loss
  - Best audiological test for acoustic neuroma
  - Useful in infants and uncooperative patients
  - Helps differentiate between cochlear from retrocochlear hearing loss

RECRUITMENT: TEST FOR COCHLEAR PATHOLOGY

00:45:38

- It is a phenomenon where there is abnormal growth in sound intensity (low intensity sound perceived as loud)
- Seen in Cochlear pathology
- Type: Subjective test (Patient gives response).

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SISI (Short Increment Sensitivity Index)

- Test for Recruitment (Cochlear pathology)
- Method:
  - Sound intensity increased/decreased by 1 dB every 5 seconds
  - Normal: Cannot appreciate the small change
  - Cochlear Pathology: Can appreciate the small change (Recruitment present)

Threshold Tone Decay Test

- Test for Retrocochlear pathology.
- Type: Subjective test.
- Method:
  - Sound stimulus given continuously
  - Interpretation:
  - Normal: Heard for 60 seconds or more.
  - Retrocochlear Pathology: Nerve undergoes fatigue → Heard for < 60 seconds.

Summary: Subjective Vs Objective Tests

Pathology	Subjective Tests	Objective Tests
Cochlear Pathology	• Pure tone audiometry • Speech audiometry • Recruitment (SISI)	• OAE • Electrocochleography
Retrocochlear Pathology	• Speech discrimination score • Tone decay test	• BERA (Very important) • Auditory steady -state response (ASSR)

APPROACH TO A PATIENT WITH HEARING LOSS

00:49:49

- Patient has hearing loss (No pain / No discharge / intact Tympanic membrane)

Step 1: Tuning Fork Tests

- Rinne test
  - AC > BC → Rinne positive → Normal or SNHL
  - AC < BC → Rinne negative → Conductive hearing loss
- Weber test
  - Lateralizes to diseased ear → Conductive
  - Lateralizes to opposite ear → SNHL
- Absolute bone conduction test → differentiate conductive vs sensorineural hearing loss

Step 2: Pta

- Confirms type of hearing loss
- Quantifies degree of loss
- If conductive Hearing loss (CHL) → Tympanometry
  - Assesses middle ear pressure & compliance
  - Normal TM + CHL > middle ear pathology
- If SNHL present → BERA / OAE
  - Otoacoustic Emissions (OAE): Identifies Cochlear pathology.
  - BERA: Identifies Retrocochlear pathology.

High Frequency Audiometry

- Standard Audiometry: Tests 250 Hz to 8000 Hz (Normal hearing 20-20,000 Hz).
- High Frequency Audiometry: Tests 16,000 Hz, 20,000 Hz, 25,000 Hz.
- **Indications:**
  - Identification of Noise-Induced Hearing Loss (NIHL).
  - Identification of Ototoxicity (ototoxic drugs).

Important Information

- Stapedial reflex is mediated by 8th nerve (Afferent) and 7th nerve (Efferent)
- Higher auditory center determine localize & discriminate
- Screening test for hearing loss is OAE
- Confirmatory test for hearing loss is BERA
- Objective tests for hearing are OAE / BERA / Electrocochleography
- Subjective test for hearing are TFT / PTA / SISI/ Tone decay test

DEGREE OF HEARING LOSS

00:53:53

WHO Grading

Category	Hearing Level (dB HL)
0-None	≤ 25 dB
1-Slight/Mild	26-40 dB
2-Moderate	41-60 dB
3-Severe	61-80 dB
4-Profound	≥ 81 dB

ASHA Grading (American Speech-Language-Hearing Association)

Category	Hearing Level (dB HL)
Mild	26-40 dB
Moderate	41-55 dB
Moderately Severe	56-70 dB
Severe	71-90 dB
Profound	> 91 dB

Approximate Hearing Loss by Condition

Pathology	Approximate hearing loss
Complete obstruction of the ear canal	40 dB
Perforation of TM	10-40 dB
Ossicular interruption with intact TM	54 dB
Ossicular interruption with perforated TM	38 dB (lesser than above because some sound reaching directly to the footplate through the perforation)
Complete fixation of footplate	60 dB

ERROR FILES

00:55:41

- **False Negative Rinne:** Seen in Unilateral Severe to Profound SNHL.
- **Audiogram Shapes:**
  - **Down-sloping:** Presbycusis.
  - **Up-sloping:** Meniere's Disease.
- **Test Specificity Confusions:**
  - Subjective for Cochlear → SISI
  - Objective for Cochlear → OAE, ECochG
  - Subjective for Retrocochlear → Tone Decay.
  - Objective for Retrocochlear → BERA.

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PYQ

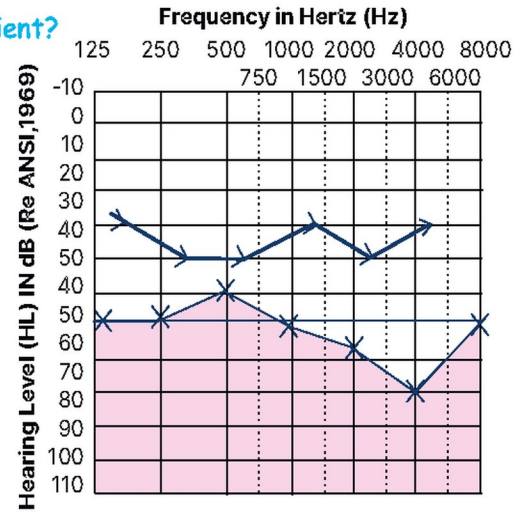
00:56:46

Patient had negative Rinne on 256 and 512 Hz tuning fork but positive Rinne on 1024 Hz. What is the air-bone gap?

a. 15 to 30 dB
b. 30 to 45 dB
c. 45 to 60 dB
d. 60 dB

Ans. B

Q. What is the interpretation of the following PTA conducted on a patient?



a. Mixed hearing loss
b. Sensorineural hearing loss
c. Conductive hearing loss
d. Auditory neuropathy spectrum disorder

Ans. B

Q. Carhart Notch is seen in which disease?

a. Otosclerosis
b. Meniere's disease
c. CHL
d. SCC dehiscence

Ans. A

Q. M/c test used in screening of neonatal hearing loss?

a. OAE
b. BERA
c. PTA
d. None of the above

Ans. A

Q. Which of the following conditions show maximum hearing loss?

a. Serous otitis media
b. Partial fixation of footplate of stapes
c. Ossicular discontinuity with intact TM
d. Ossicular discontinuity with perforated TM

Ans. C

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4. ASSESSMENT OF VESTIBULAR SYSTEM

NYSTAGMUS

00:00:55

Approach

1. Central vs Peripheral component of nystagmus
2. Rule / Direction of nystagmus (what it indicates)
3. How to evoke nystagmus (how to stimulate the labyrinth)

3 Ways To Evoke Nystagmus

- Temperature change → Caloric test
- Pressure change → Fistula test
- Position change → Dix-Hallpike test

Central Vs Peripheral Nystagmus

00:02:45

Feature	Peripheral (Labyrinthine)	Central (Brain)
Latency	Present • Problem in balance receptors: macula + crista • Utricle/saccule/semicircular canals involved → vertigo	Absent
Fatigability	Fatigable (d/t repeated stimulation)	Not fatigable
Duration	Limited (1.5 min to 2 min)	Long-lasting
Direction	Usually fixed	Direction-changing / random
Optic fixation	Disappears	Does NOT disappear
Torsion with horizontal/ Vertical	Present	Pure Horizontal / pure vertical /pure torsional

- Latency = time from stimulation to appearance of nystagmus
- If you stimulate at 0 sec and nystagmus comes at 10 sec → latency present
- Peripheral (labyrinthine): Latency present
  - Reason: labyrinth has to be stimulated → impulses travel via vestibular nerve → vestibular nucleus → MLF → $3^{rd}/4^{th}/6^{th}$ nuclei → then nystagmus appears
- Central: No latency
  - From stimulation to nystagmus → immediately

Direction Of Nystagmus

00:07:00

- Normal eyes stay in midline because: Equal pull from both labyrinths → midline
- "Direction of nystagmus" means direction of the FAST component
  - Nystagmus has fast + slow component
  - Direction is named by fast component
  - Example: Fast to right + slow to left → direction is right

Type of Lesion	Seen in which condition	Direction of nystagmus
Hyperactive lesion of labyrinth	BPPV	Toward more active side → i.e., Toward the involved ear
Hypoactive lesion of labyrinth	Vestibular neuritis, purulent labyrinthitis	Toward more active side → i.e., Toward the normal ear

- **For example:**
  - In right PSC BPPV - Vertical nystagmus toward right
  - Right vestibular neuritis - Horizontal nystagmus to left (active) side.

Direction Based On Which Canal Is Involved

00:10:00

INICET 2025

Due to Involvement of	Type of Peripheral Origin Nystagmus seen	Seen in which conditions
Vertical canal (PSC, SSC)	Vertical nystagmus with torsion	Posterior and superior canal BPPV, Superior SCC Dehiscence
Superior canal	Down beating	
Posterior canal	Upbeating	
Horizontal SCC	Horizontal nystagmus	Horizontal SCC, BPPV
Complete involvement of labyrinth (hypoactive)	Horizontal nystagmus with torsion	Vestibular neuritis (Complete destruction of the vestibular nerve), purulent labyrinthitis, late stages of Meniere's

Evoke Nystagmus

Position Change → Dix-hallpike Test

00:12:37

AIIMS 2020

- Investigation of choice for BBPV
- Benign paroxysmal positional vertigo occurs due to displacement of small crystals of calcium carbonate (otoliths) in semicircular canal area
  - Vertigo in BBPV lasts a few seconds to few minutes.
- Principle of the test
- Normally, the otoliths are present in the ampulated end of the semi-circular canals.
- Putting the patients head in low position, displaces the otoliths and provokes vertigo and nystagmus

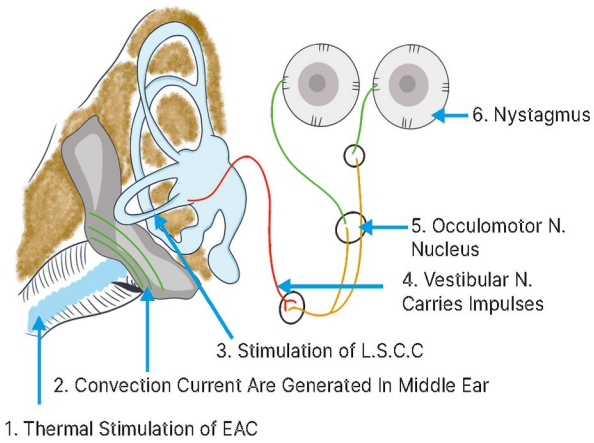
Inner ear canal involved	Most common?	Primary Diagnostic Test	What happens in the test
Posterior canal	Most common (~85-90%)	Dix-Hallpike test	Vertigo + upward & twisting eye movements (torsion)
Horizontal (lateral) canal	Less common	Supine Roll Test	Vertigo when head turned left/right while lying flat
Anterior (superior) canal	Rare	Deep Head hanging test	Vertigo when head is deeply extended

Thermal Changes- Caloric Test

00:16:00

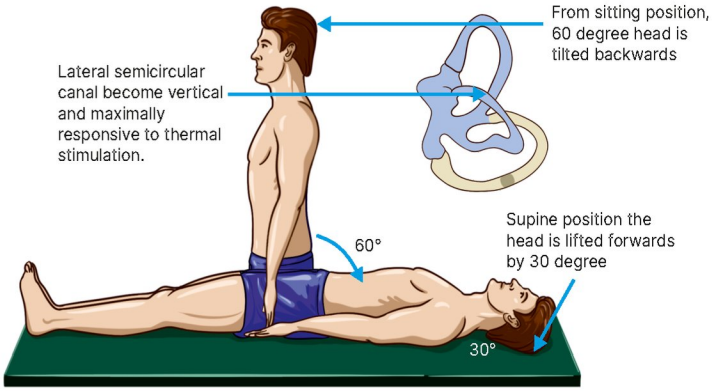
Principle

We irrigate warm or cold water in external auditory canal
↓
Stimulation of lateral semicircular canal in the medial wall
↓
Stimulation of Superior vestibular nerve
↓
Vestibular nerve
↓
Vestibular nucleus
↓
$3^{rd}/4^{th}/6^{th}$ nuclei
↓
Nystagmus



Position

- If supine → head up 30 degrees
- If sitting → head down 60 degrees



Procedure And Inference

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- It is a test for VOR pathway
- Unilateral testing done using caloric test (each labyrinth cane be tested)
- Temperature- Bi-thermal (Fitzgerald/Hallpike test)- 30°C to 44°C
- As body temp is 37°C, 7° below and 7° above is used
- Semicircular canal stimulted- Lateral canal
- Normal response: COWS
  - Cold water → nystagmus to Opposite side
  - Warm water → nystagmus to Same side
  - Example: Right warm → nystagmus to right; Right cold → nystagmus to left
- If no response / no nystagmus
- Hypoactive labyrinth OR
- Vestibular nerve/pathway problem
- Inner ear is not functioning

Exceptions

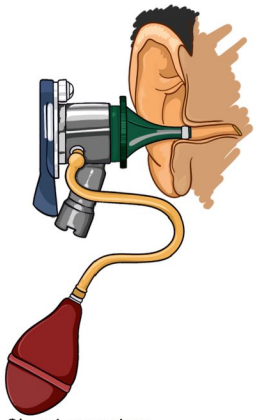
- Tympanic membrane perforation
  - Do NOT irrigate water → Instead use air → Dundas Grant tube test
- If no pinna and no canal →Barany's rotational chair test
  - Patient sits on rotating chair with VR glasses
  - Rotation stimulates crista (angular/rotational motion)
  - VR glasses record nystagmus

YOURWISH

Pressure Change → Fistula Test

00:21:30

- Pressure stimulation of the EAC leading to labyrinthine stimulation
- Done by pressure on tragus or siegelization
  - Siegle's pneumatic speculum → inflate/deflate bulb → increase/decrease canal pressure
- Vertigo increases on:
  - Cough
  - Sneeze
  - Lifting heavy weight (because middle ear pressure increases)
- Normally there is no direct communication between middle ear and inner ear. So pressure is not transmitted to inner ear. Fistula test should be NEGATIVE in normal person



Siegel speculum

- In case of lateral semicircular canal dehiscence/Oval window rupture/Round window rupture, there is a direct communication between middle ear and inner ear. Pressure is transmitted, stimulating vestibular nerve and nucleus resulting in nystagmus

Inference

- Positive Fistula test- Communication present
- Negative Fistula test- In normal individuals
- False Positive Fistula test- No communication, but test is positive
- False negative fistula test- Communication is present, but test is negative

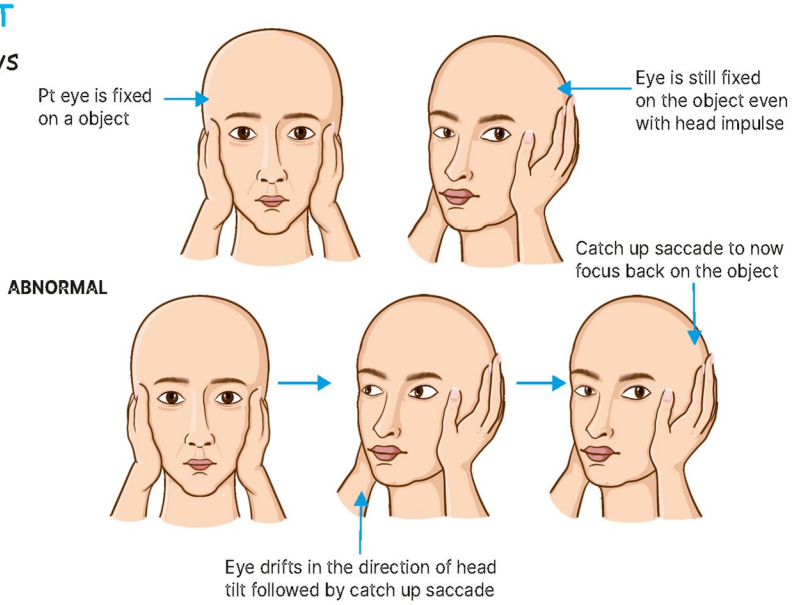
Fistula test result	Seen in which conditions
Positive (vertigo and nystagmus on doing the test)	Fistula on medial wall e.g., perilymphatic fistula, LSC erosion, fenestration operation
False positive / Hennebert sign (test +ve without fistula on medial wall)	• Meniere's • Hypermobility footplate (congenital syphilis) SCC dehiscence
False negative (test -ve in spite of fistula on medial wall)	• Dead labyrinth • Cholesteatoma covering fistula

HEAD IMPULSE TEST

00:27:30

Neet PG 2024

- Test for VOR pathways



YOURWISH

Procedure

- Ask patient to fixate eyes on a target (examiner's nose/finger/pen)
- Examiner suddenly rotates head to right/left

Interpretation

- Normal: Eye remains fixed on object
- Abnormal
- Eye deflects with head movement then comes back to catch the object: This is a saccade
- Saccade present = labyrinthine / inner ear abnormality
- Abnormal on right thrust → right vestibular hypofunction

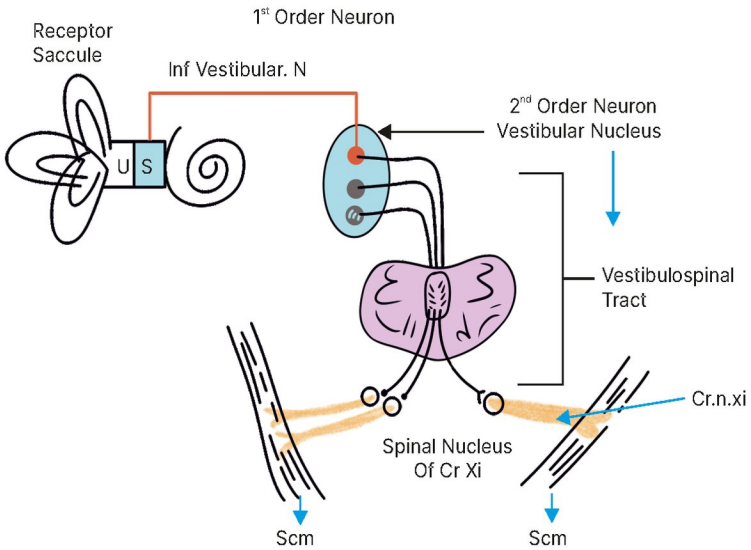
VEMP (VESTIBULAR EVOKED MYOGENIC POTENTIAL)

00:29:30

- Now this is for vestibulo-spinal reflex, not VOR
- VEMP = vestibular system stimulated → muscle contraction recorded by electrodes

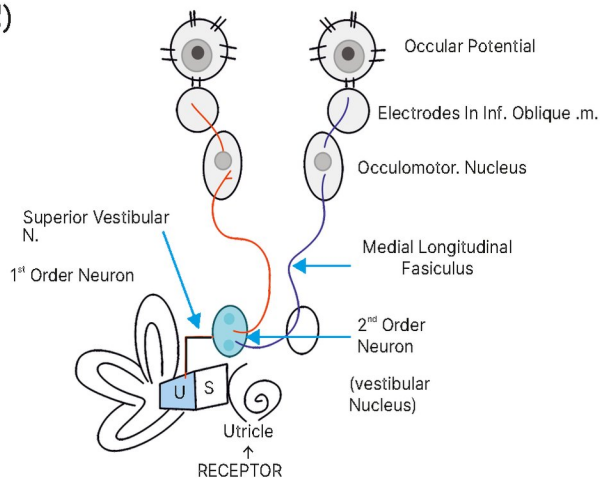
cVEMP (cervical VEMP)

- Electrode on sternocleidomastoid
- Tests inferior vestibular nerve
- Pathway described: saccule → inferior vestibular nerve → vestibular nucleus → spinal cord → spinal accessory nerve → SCM contraction



oVEMP (ocular VEMP)

- Records ocular muscle contraction (inferior oblique mentioned)
- stimulates utricle
- Impulses via superior vestibular nerve



VEMP TEST RESULT	CONDITIONS
Hyperactive response, these patients have vertigo on loud sounds (Tullio's phenomenon)	• Meniere's • fistula • canal dehiscence • Hypermobile footplate (Congenital syphilis)
Hypoactive response	• labyrinthitis • vestibular neuritis • acoustic neuroma

Common Student Errors

- Caloric test positive- Normal
- Fistula test positive- Abnormal Dix-Hallpike test positive- BBPV
- cVEMP vs oVEMP confusion
- cVEMP → cervical muscle (SCM) → inferior vestibular nerve
- oVEMP → ocular muscle → superior vestibular nerve
- Peripheral vs central nystagmus confusion
- Peripheral: Latency present; Fatigable; Limited duration
- Central: No latency; Not fatigable; Unlimited / direction changing

MCQs

00:34:10

Q. A patient is found to have acute vertigo and horizontal nystagmus with the slow component toward the left. In which of the following conditions would you see this finding?

A. Posterior canal BPPV

B. Superior canal BPPV

C. Right hypoactive labyrinth

D. Left hypoactive labyrinth

Ans. D

Explanation:

- Low component to left → Fast component to right → Direction of nystagmus is right
  - Rigth is hyperactive or left is hypoactive
- Posterior and superior BBPV- Vertical nystagmus

Q. Head Impulse Test is primarily used to assess

A. Otolith function

B. Central vestibular pathway

C. Vestibulo-ocular reflex

D. Cochlear reserve

Ans. C

YOURWISH

5. ACOUSTIC NEUROMA

- Benign tumor
- Arises from the internal auditory canal (IAC)
- Extends toward the cerebellopontine angle (CPA)
- MC tumor of CPA → Acoustic neuroma
- Accounts for ~80% of CPA tumors
- Cell of origin → Schwann cells

Anatomy – Internal Auditory Canal (Iac)

00:00:17

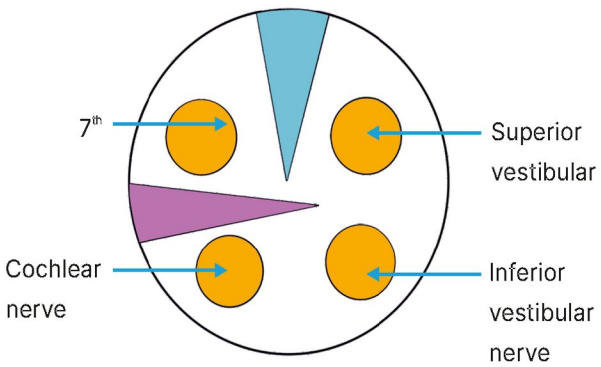
- **Structures passing through IAC**
  - Facial nerve (CN VII)
  - Vestibulocochlear nerve (CN VIII)
- **Branches of Vestibulocochlear Nerve**
  - Cochlear nerve
  - Vestibular nerve
- **Branches of the Vestibular nerve**
  - Superior vestibular nerve
  - Inferior vestibular nerve

Quadrant Arrangement In Iac

- Two bony crests divide the internal auditory canal.
  - **Bill's bar-** Vertical crest
  - **Falciform crest-** Horizontal crest

Nerve Arrangement In Quadrants

- Anterosuperior quadrant
  - Facial nerve (CN VII)
- Anteroinferior quadrant
  - Cochlear nerve
- Posterosuperior quadrant
  - Superior vestibular nerve
- Posteroinferior quadrant
  - Inferior vestibular nerve- MC site of origin of Acoustic neuroma

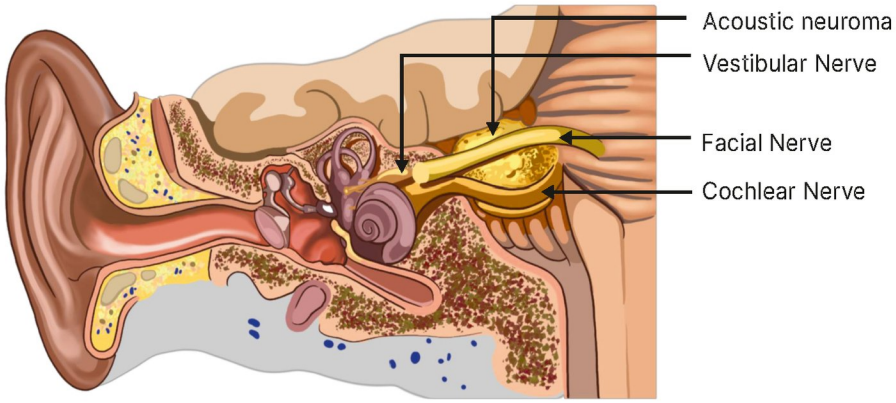


Pathology

- Benign tumor, locally invasive
- Originates from Schwann cells of the vestibular nerve, but sometimes even from the Cochlear nerve
- As it expands, it causes widening of the canal and appears in the CP angle

YOURWISH

FMGE 2021



Acoustic neuroma on vestibulocochlear nerve puts pressure on facial nerve

Clinical Features

00:07:00

NEET PG 2023

- Symptoms occur due to compression of cranial nerves

Early Stage – Internal Auditory Canal

Vestibular nerve involvement	• Vertigo • Imbalance • Often compensated by the opposite vestibular system
Cochlear nerve involvement	• Retrocochlear Sensorineural hearing loss

Facial Nerve Involvement	• Early involvement: Affects sensory fibers
	  ○ Feature: Loss of sensation over the concha
	  ○ Hitzelberger's Sign: Loss of conchal sensation (indicates facial nerve sensory involvement)
	• Late involvement: LMN facial palsy

Late Stage – Expanding Tumor

- As the tumor enlarges, additional cranial nerves get compressed

Direction	Cranial Nerves Involved	Symptoms
Superior tumor expansion	CN V (Trigeminal) CN VI (Abducens)	Loss of corneal reflex Lateral rectus palsy
Inferior tumor expansion	CN IX CN X CN XI	Dysphagia Speech difficulty
Medial tumor expansion	Brainstem Cerebellum	Ataxia Raised intracranial pressure

Key Clinical Indications

- Most common age: 40–60 years
- Bilateral tumors are part of neurofibromatosis type 2

- Unilateral sensorineural hearing loss
- Poor speech discrimination
- Cranial nerve involvement
- Brainstem/cerebellum involvement

Audiometric Findings

00:09:00

Tuning Fork Tests

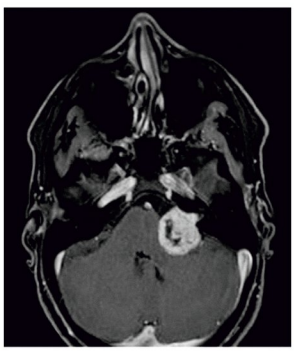
- **Rinne test-** Positive
  - False positive in severe to profound U/L SNHL (>70-90 dB)
- **Weber test-** Lateralizes to the normal ear
- **Absolute bone conduction-** Shortened
- Acoustic reflex delay- Positive
- Speech discrimination- Poor
- Tone Decay Test- Positive
  - Patient hears the tone for less than 60 seconds

Audiological Investigation Of Choice

- BERA (Brainstem Evoked Response Audiometry)
- Delay in Wave 5

General Investigation Of Choice

- MRI with gadolinium contrast- Ice Cream Cone Sign



Biopsy Findings

- Antoni A cells
- Antoni B cells
- Verocay bodies

Treatment

00:11:10

- Management depends on tumor size

Tumor <3 cm	• Radiosurgery- Single high-dose radiation to shrink the tumor
Tumor >3 cm	• Surgical excision Surgical approaches • Middle cranial fossa approach • Retrosigmoid approach • Transcochlear approach
NF2 with Acoustic neuroma	• Surgery may cause permanent sensorineural hearing loss • Hearing rehabilitation is achieved with the placement of an auditory brainstem implant in the lateral recess of the 4th ventricle

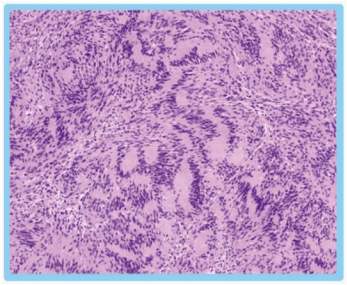
MCQs

00:13:45

Q. A young man presented with hearing loss and tinnitus. The histology image is shown below. What is the diagnosis?

INICET July 2021

A. Neurofibroma
B. Schwannoma
C. Leiomyoma
D. Rhabdomyoma



Ans: B